

A COUNTEREXAMPLE TO TATE BLUESHIFT FOR E_∞ -RINGS

ANONYMOUS

ABSTRACT. At the prime 2, we construct a connective E_∞ -ring C such that

$$K(1) \wedge C \simeq 0 \quad \text{and} \quad H\mathbb{Q} \wedge C^{tC_2} \not\simeq 0.$$

Thus $K(1)$ -acyclicity of a connective E_∞ -ring does not imply rational acyclicity of its C_2 -Tate construction. The example is the versal $ku_{(2)}$ -algebra obtained by adjoining a degree-4 class b and a nullhomotopy of $2b - u^2$, where u is the Bott class.

The proof uses an affine weight filtration whose layers are extended powers of the mod-2 Moore spectrum. McClure's finite-coefficient Bockstein calculation, followed by a Milnor inverse-limit argument, bounds the denominators detected by completed periodic K -theory. A Laurent-series model for the Tate constructions of the finite Postnikov sections then gives, for every $m \geq 0$,

$$\text{ord}\left(1 \in \pi_0((\tau_{\leq 4m} C)^{tC_2})\right) \geq 2^{s_2(m)+1}.$$

Here $s_2(m)$ is the binary digit sum of m . These orders are unbounded, which implies that C^{tC_2} is not rationally acyclic.

1. INTRODUCTION

Let p be a prime and let $n \geq 1$. Choose a finite type- n spectrum with a v_n -self map, and let $T(n)$ denote its telescope. The cyclic Tate blueshift conjecture predicts that every E_∞ -ring R satisfies

$$(1) \quad T(n) \wedge R \simeq 0 \implies T(n-1) \wedge R^{tC_p} \simeq 0.$$

The corresponding genuine-equivariant statement is [BSY22, Conjecture 9.9]. To specialize it, regard an ordinary E_∞ -ring with trivial C_p -action as its associated Borel-complete genuine C_p -ring. In the notation of that conjecture, take the ambient finite abelian group to be C_p and the family to be $\mathcal{F} = \{e\}$. Then

$$\text{cork}_p(\mathcal{F}) = 1, \quad \Phi^{\mathcal{F}} R \simeq R^{tC_p}.$$

Thus (1) is its C_p specialization. At height one, $T(1)$ and $K(1)$ have the same Bousfield class, while $T(0) = K(0) = H\mathbb{Q}$. We disprove (1) for $(p, n) = (2, 1)$.

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From now on spectra are localized at 2 unless explicitly indicated otherwise, and group actions are trivial unless specified otherwise. For a spectrum X , write $X_{\mathbb{Q}} = H\mathbb{Q} \wedge X$ and $\tau_{\leq N}X$ for its Postnikov truncation. Let

$$A = ku_{(2)}, \quad \pi_* A = \mathbb{Z}_{(2)}[u], \quad |u| = 2.$$

Here u is the Bott class. Write CAlg_A for the ∞ -category of commutative A -algebras. Tensor products are relative tensor products in the indicated ambient category; in CAlg_A they are pushouts. For an A -module V , write $\mathrm{Sym}_A(V)$ for the free commutative A -algebra on V . The subscripts b and w below name the canonical degree-4 generators of the corresponding free algebras.

Theorem 1.1. *Let C be the pushout in CAlg_A*

$$(2) \quad C = \mathrm{Sym}_A(\Sigma^4 A)_b \otimes_{\mathrm{Sym}_A(\Sigma^4 A)_w} A,$$

where the two structure maps are classified by

$$w \mapsto 2b - u^2, \quad w \mapsto 0.$$

Then C is connective and

$$K(1) \wedge C \simeq 0, \quad H\mathbb{Q} \wedge C^{tC_2} \not\simeq 0.$$

Consequently, (1) has a negative answer.

Theorem 1.1 disproves [BSY22, Conjecture 9.9] as stated in arXiv version 1.

The first assertion follows directly from the defining relation. After smashing with $K(1)$, multiplication by 2 is null and multiplication by u is invertible; the nullhomotopy of $2b - u^2$ therefore makes an invertible endomorphism null. Rationally, the change of variables

$$b \mapsto \frac{w + u^2}{2}$$

cancels the generator–relation pair and gives $C_{\mathbb{Q}} \simeq H\mathbb{Q}[u]$. It remains to control the integral lattice in this rational polynomial ring.

We filter C by affine weight. Its k th filtration layer is

$$\Sigma^{4k} A \wedge D_k(S/2),$$

where D_k is the k th extended power. McClure’s calculation in [BMMS86, Chapter IX] controls the contribution of each layer. We apply that calculation before passing to the inverse limit; a separate Milnor argument produces the completed periodic K -theory detector used below. Iterating the resulting layer bounds shows that the denominator exponent in degree $4m$ is at most $2m - s_2(m)$, where $s_2(m)$ is the sum of the binary digits of m .

Each finite Postnikov section of C is perfect as an A -module. The projection formula therefore reduces its Tate construction to a two-term Laurent-series model for A^{tC_2} . A Smith-normal-form calculation gives

$$\mathrm{ord}(1 \in \pi_0((\tau_{\leq 4m} C)^{tC_2})) \geq 2^{s_2(m)+1}.$$

Taking $m = 2^q - 1$ makes these orders unbounded. Since the maps

$$C^{tC_2} \longrightarrow (\tau_{\leq N} C)^{tC_2},$$

preserve the Tate unit, this forces $H\mathbb{Q} \wedge C^{tC_2}$ to be nonzero. The last step uses only these unital maps; it does not require an inverse-limit comparison.

Section 2 establishes the basic properties of C . Section 3 constructs the affine weight filtration, and Section 4 proves the periodic denominator estimate. Section 5 gives the Tate model for finite Postnikov sections. Section 6 derives the unbounded unit orders and proves Theorem 1.1.

2. THE VERSAL ALGEBRA

The next proposition establishes connectivity, identifies $C_{\mathbb{Q}}$, and proves $K(1)$ -acyclicity.

Proposition 2.1. *The E_{∞} -ring C is connective, and*

$$C_{\mathbb{Q}} \simeq A_{\mathbb{Q}} \simeq H\mathbb{Q}[u].$$

Moreover, $K(1) \wedge C \simeq 0$.

Proof. Connectivity follows because free commutative algebras on connective A -modules and pushouts of connective E_{∞} - A -algebras are connective. After rationalization, the map

$$\mathrm{Sym}_{A_{\mathbb{Q}}}(\Sigma^4 A_{\mathbb{Q}})_w \longrightarrow \mathrm{Sym}_{A_{\mathbb{Q}}}(\Sigma^4 A_{\mathbb{Q}})_b, \quad w \longmapsto 2b - u^2,$$

is an equivalence. Its inverse sends b to $(w + u^2)/2$. Taking the pushout gives $C_{\mathbb{Q}} \simeq A_{\mathbb{Q}}$.

For the final assertion, write

$$K(1) \wedge C \simeq (K(1) \wedge A) \otimes_A C.$$

Multiplication by u is an equivalence on $K(1) \wedge A$. For example, the cofiber of $u: \Sigma^2 A \rightarrow A$ is $H\mathbb{Z}_{(2)}$, which is $K(1)$ -acyclic. Hence u also acts invertibly on $K(1) \wedge C$. In C the chosen cell gives a homotopy $u^2 \simeq 2b$. After tensoring with $K(1)$, its right-hand side is null because $2 = 0$ on every $K(1)$ -module. Thus the invertible endomorphism u^2 of $K(1) \wedge C$ is null, forcing $K(1) \wedge C \simeq 0$. $\square$

3. THE AFFINE WEIGHT FILTRATION

We construct the affine weight filtration in the stable ∞ -category of filtered A -modules. This ensures that its associated graded computes the derived pushout defining C .

For $k \geq 0$, an A -module X , and a spectrum Y , put

$$D_k^A(X) = (X^{\otimes_A k})_{h\Sigma_k}, \quad D_k(Y) = (Y^{\wedge k})_{h\Sigma_k}.$$

The symmetric group acts by permutation. For $a \geq 1$, we use the suspension convention $S/2^a = \mathrm{cofib}(2^a: S \rightarrow S)$ (thus this is the suspension of the Moore spectrum M_a used in BMMS) and put $A/2 = A \wedge S/2$. For a filtered module $F_* X$, put $\mathrm{gr}_k^F X = \mathrm{cofib}(F_{k-1} X \rightarrow F_k X)$.

Proposition 3.1 (Affine weight filtration). *There is an increasing multiplicative filtration*

$$0 = F_{-1}C \longrightarrow F_0C \longrightarrow F_1C \longrightarrow \cdots$$

of the underlying A -module of C such that:

- (i) $\operatorname{colim}_k F_k C \simeq C$ and $F_0 C \simeq A$;
 - (ii) for every $k \geq 0$,
- (3) $\operatorname{gr}_k^F C \simeq D_k^A(\Sigma^4 A/2) \simeq \Sigma^{4k} A \wedge D_k(S/2)$;
- (iii) for every $m \geq 0$, the cofiber $\operatorname{cofib}(F_m C \rightarrow C)$ is $4(m+1)$ -connective. In particular, for every $r \geq 0$,
- (4) $\pi_{2r} F_{\lfloor r/2 \rfloor} C \xrightarrow{\cong} \pi_{2r} C$;
- (iv) every $\pi_i C$ is a finitely generated $\mathbb{Z}_{(2)}$ -module.

Proof. Let $V = \Sigma^4 A$ and $W = \Sigma^4 A$. The category

$$\operatorname{Fil}(\operatorname{Mod}_A) = \operatorname{Fun}(\mathbb{N}, \operatorname{Mod}_A)$$

is symmetric monoidal under Day convolution:

$$(X \otimes_{\operatorname{Day}} Y)_n = \operatorname{colim}_{i+j \leq n} X_i \otimes_A Y_j.$$

For an A -module X , write $X\{d\}$ for the step filtration that is zero below d and equal to X from degree d onward. Form the filtered commutative algebras

$$\mathcal{B} = \operatorname{Sym}_{\operatorname{Fil}(\operatorname{Mod}_A)}(V\{1\}), \quad \mathcal{S} = \operatorname{Sym}_{\operatorname{Fil}(\operatorname{Mod}_A)}(W\{1\}), \quad \mathcal{A} = A\{0\}.$$

The class $2b - u^2 \in \pi_4 \mathcal{B}_1$ defines a filtered algebra map $\mathcal{S} \rightarrow \mathcal{B}$: its two summands have filtration degrees one and zero. The zero class defines $\mathcal{S} \rightarrow \mathcal{A}$. Set

$$(5) \quad \mathcal{C} = \mathcal{B} \otimes_{\mathcal{S}} \mathcal{A} \quad \text{in } \operatorname{CAlg}(\operatorname{Fil}(\operatorname{Mod}_A)), \quad F_n C = \mathcal{C}_n.$$

The colimit functor from filtered to ordinary A -modules is strong symmetric monoidal for Day convolution and preserves colimits. It carries (5) to (2). Evaluation in filtration degree zero gives $F_0 C \simeq A$. Day convolution also provides the pairings $F_i C \otimes_A F_j C \rightarrow F_{i+j} C$.

Let $\operatorname{Gr}(\operatorname{Mod}_A)$ be graded A -modules with convolution, and define

$$(\operatorname{gr} X)_0 = X_0, \quad (\operatorname{gr} X)_n = \operatorname{cofib}(X_{n-1} \rightarrow X_n) \quad (n > 0).$$

In a stable category, the total cofiber formula for the boundary diagram in Day convolution gives

$$\operatorname{gr}_n(X \otimes_{\operatorname{Day}} Y) \simeq \bigoplus_{i+j=n} \operatorname{gr}_i X \otimes_A \operatorname{gr}_j Y.$$

Thus gr is colimit preserving and strong symmetric monoidal. One may equivalently verify this on the two-sided Day-convolution bar construction for (5). Applying gr to that pushout gives

$$\text{gr } \mathcal{C} \simeq \text{gr } \mathcal{B} \otimes_{\text{gr } \mathcal{S}} \text{gr } \mathcal{A}.$$

The initial form of $2b - u^2$ in filtration one is $2b$; the term u^2 has filtration zero. Therefore the associated graded of the attaching diagram is induced by the following maps, where brackets denote weight:

$$V[1] \xleftarrow{2} W[1] \longrightarrow 0.$$

Since the free commutative algebra functor preserves colimits,

$$\begin{aligned} \text{gr } \mathcal{C} &\simeq \text{Sym}_A(V[1]) \otimes_{\text{Sym}_A(W[1])} A \\ &\simeq \text{Sym}_A((\Sigma^4 A/2)[1]). \end{aligned}$$

Its weight- k summand is $D_k^A(\Sigma^4 A/2)$.

The second equivalence in (3) requires an equivariant suspension calculation. If ρ_k is the real permutation representation of Σ_k , then

$$(\Sigma^4 S/2)^{\wedge k} \simeq S^{4\rho_k} \wedge (S/2)^{\wedge k}$$

as naive Σ_k -spectra. The representation $4\rho_k$ is the underlying real representation of the complex representation $2\rho_{k,\mathbb{C}}$, of complex rank $2k$. The complex orientation of $A = ku_{(2)}$ supplies the Borel Thom equivalence

$$(6) \quad A \wedge (S^{4\rho_k} \wedge X)_{h\Sigma_k} \simeq \Sigma^{4k} A \wedge X_{h\Sigma_k}$$

for every naive Σ_k -spectrum X : the left-hand side is the Thom spectrum, over $X_{h\Sigma_k}$, of the pullback of the complex bundle $2\rho_{k,\mathbb{C}}$. Applying this equivalence to $X = (S/2)^{\wedge k}$ proves (3).

The spectrum $D_k(S/2)$ is connective, so $\text{gr}_k^F C$ is $4k$ -connective. For $n > m$, the cofiber of $F_m C \rightarrow F_n C$ is an iterated extension of layers with $k > m$, each $4(m+1)$ -connective. Passing to the filtered colimit shows that $\text{cofib}(F_m C \rightarrow C)$ has the same connectivity. The resulting long exact sequence gives (4).

Finally, $(S/2)^{\wedge k}$ is finite and connective. The homotopy orbit spectral sequence shows that $D_k(S/2)$ is of finite type, and the connective Atiyah–Hirzebruch spectral sequence then shows that every $\pi_i(A \wedge D_k(S/2))$ is a finitely generated $\mathbb{Z}_{(2)}$ -module. The filtration is degreewise eventually constant, so the same holds for C . $\square$

Remark 3.2. The objects $F_m C$ are filtered pieces of the underlying A -module; they need not be commutative A -subalgebras. The construction gives the multiplicative pairings $F_i C \otimes_A F_j C \rightarrow F_{i+j} C$, which is all that is used below. The extended-power layers in (3) are precisely what the naive ring $\mathbb{Z}_{(2)}[u, b]/(2b - u^2)$ omits.

4. A COMPLETED PERIODIC K -THEORY ESTIMATE

The input from [BMMS86, Chapter IX] is a finite-coefficient calculation used below to derive an estimate for the completed functor. Put $KU_{(2)} = A[1/u]$, write $KU_2^\wedge$ for its 2-completion, and let $\mathbb{Z}_2$ be the 2-adic integers and $\mathbb{Q}_2 = \mathbb{Z}_2[1/2]$. For an A -module X , define

$$(7) \quad \mathcal{K}(X) = \operatorname{holim}_{s \geq 1} (KU_{(2)} \otimes_A X \wedge S/2^s),$$

where the tower uses the coefficient-reduction maps. This is an exact functor: every functor at finite s is exact, and limits are exact in a stable category. Notice that

$$\mathcal{K}(A) \simeq KU_2^\wedge.$$

For a nonnegative integer m , let $s_2(m)$ denote the sum of its binary digits.

Proposition 4.1 (McClure's extended-power estimate). *For $k \geq 1$, let $\nu_2(k)$ be the 2-adic valuation and put $h_k = 1 + \nu_2(k)$. For every $q \in \mathbb{Z}$,*

$$(8) \quad \pi_{2q} \mathcal{K}(A \wedge D_k(S/2)) \cong \mathbb{Z}/2^{h_k}, \quad \pi_{2q+1} \mathcal{K}(A \wedge D_k(S/2)) = 0.$$

In particular, all these homotopy groups are annihilated by 2^{h_k} .

Proof. We spell out the specialization of [BMMS86, Chapter IX]. The passage to coefficient completion is a separate inverse-limit step. In the notation of that source,

$$M_a = S^{-1} \cup_{2^a} e^0, \quad E_a^{\text{BMMS}} = E \wedge \Sigma M_a, \quad E_n(X; a) = \pi_n(E_a^{\text{BMMS}} \wedge X).$$

Thus our convention $S/2^a = \Sigma M_a$ is the suspension used in their coefficient theory. The Moore subbasis class for M_1 is even; its suspension therefore gives a one-element *odd* subbasis $\{z\}$ for our $S/2$ [BMMS86, Chapter IX, p. 295 and the paragraph following Lemma 4.3, p. 317]; see also [BMMS86, Chapter IX, Definition 3.8 and Proposition 3.9, pp. 313–314]. On the s th page of the coefficient K -theory Bockstein spectral sequence, [BMMS86, Chapter IX, Theorem 3.12, p. 316] gives generators, in terms of the operators π , R , and β_j used there,

$$a_s = \pi^{s-1} R^{s-1} z, \quad b_j = \pi^j \beta_{j+1} R^j z \quad (j \geq s-1).$$

Here a_s is odd of weight 2^{s-1} , while b_j is even of weight 2^j . Theorem IX.4.1 and Remark IX.4.2 identify the weight- k summand with the Bockstein spectral sequence for $D_k(S/2)$ [BMMS86, Chapter IX, Theorem 4.1 and Remark 4.2, p. 317]. The additive basis in Remark IX.3.11(ii) has the following consequence: the odd indecomposable a_s occurs at most once, and in the height-one specialization the additive quotient in Theorem IX.3.12 has exponent < 2 for each b_j [BMMS86, Chapter IX, Remark 3.11(ii), p. 315 and Theorem 3.12, p. 316]. Hence the weight-generating series on page s is

$$(9) \quad (1 + t^{2^{s-1}}) \prod_{j \geq s-1} (1 + t^{2^j}) = \frac{1 + t^{2^{s-1}}}{1 - t^{2^{s-1}}} = 1 + 2 \sum_{q \geq 1} t^{q2^{s-1}}.$$

Thus a positive weight k contributes one class in each parity on page s exactly when 2^{s-1} divides k . Its Bockstein height is therefore $h_k = 1 + \nu_2(k)$. More explicitly, when $k = q2^{s-1} > 0$, the binary expansion of q specifies the unique even basis monomial, which we denote by $m(q)$, while $a_s m(q-1)$ is the unique odd one. At the terminal page $h = h_k$, write $k = q2^{h-1}$ with q odd. The Bockstein differential rule on [BMMS86, Chapter IX, p. 313] gives

$$d_h(a_h m(q-1)) = b_{h-1} m(q-1) = m(q).$$

Thus the height- h subbasic lift is odd, and its Bockstein is the even class. For $p = 2$ and $s = 1, 2$, the cited Theorem IX.3.12 asserts only the additive description used here; no multiplicative conclusion is needed.

Proposition IX.3.9 gives, at coefficient level 2^a , cyclic groups of order $2^{\min(a, h_k)}$ in both parities [BMMS86, Chapter IX, Proposition 3.9, pp. 313–314]. To pass to (7), one must also inspect the transition maps. Choose the odd height- h_k subbasic class e , and for $a \geq h_k$ put

$$e_a = p_*^{a-h_k} e, \quad f_a = \beta_a e_a.$$

Here p_* is the coefficient-inclusion operator of BMMS. Under their coefficient-reduction operator π , the identities in [BMMS86, Chapter IX, Proposition 3.1(ii)–(iii), p. 306] give

$$e_{a+1} \mapsto 2e_a, \quad f_{a+1} \mapsto f_a.$$

The odd inverse system is therefore pro-zero, while the even system is eventually constant. Both are Mittag-Leffler and both $\lim^1$ terms vanish. The finite coefficient groups and their bonding maps are the data supplied by the cited Bockstein calculation. Applying the Milnor exact sequence to the inverse limit in (7), followed by Bott periodicity, gives (8). $\square$

Remark 4.2. Proposition 4.1 uses the inverse limit in (7). Since $D_k(S/2)$ need not be dualizable, comparison with $KU_2^\wedge \wedge D_k(S/2)$ is not available in general. The finite coefficient tower is the object to which the cited Bockstein spectral sequence applies; its inverse-limit identification is the Milnor argument above.

Apply $\mathcal{K}$ to the filtration of Proposition 3.1. By Proposition 4.1, the map

$$\mathcal{K}(F_{k-1}C)[1/2] \longrightarrow \mathcal{K}(F_kC)[1/2]$$

is an equivalence. We identify all these rationalizations with

$$(10) \quad \mathcal{K}(F_0C)[1/2] \simeq H\mathbb{Q}_2[u^{\pm 1}].$$

Let Λ_k denote the image of $\pi_* \mathcal{K}(F_kC)$ in $\mathbb{Q}_2[u^{\pm 1}]$ under this identification.

Lemma 4.3. *For every $k \geq 1$,*

$$2^{1+\nu_2(k)} \Lambda_k \subseteq \Lambda_{k-1}.$$

Consequently, for every $m \geq 0$,

$$(11) \quad \Lambda_m \subseteq 2^{-\Delta_m} \mathbb{Z}_2[u^{\pm 1}], \quad \Delta_m = \sum_{k=1}^m (1 + \nu_2(k)) = 2m - s_2(m).$$

Proof. Let $y \in \pi_*\mathcal{K}(F_k C)$. Its image in the homotopy of the cofiber $\mathcal{K}(\mathrm{gr}_k^F C)$ is killed by $2^{1+\nu_2(k)}$, by (3) and Proposition 4.1. Exactness therefore lifts that multiple of y to $\pi_*\mathcal{K}(F_{k-1}C)$. This proves the first claim after using (10).

Iteration starts from $\Lambda_0 = \mathbb{Z}_2[u^{\pm 1}]$ and gives the inclusion in (11). Finally, Legendre's formula gives

$$\sum_{k=1}^m \nu_2(k) = \nu_2(m!) = m - s_2(m).$$

□

We convert the bound (11) on the periodic lattices Λ_m into a bound on the connective rational lattice of C .

Definition 4.4. For $r \geq 0$, define $d_r \geq 0$ by

$$(12) \quad \mathrm{im}(\pi_{2r}C \longrightarrow \pi_{2r}C_{\mathbb{Q}} = \mathbb{Q}u^r) = 2^{-d_r}\mathbb{Z}_{(2)}u^r.$$

The image is a nonzero rank-one lattice by Propositions 2.1 and 3.1; it contains $\mathbb{Z}_{(2)}u^r$ through the structural map $A \rightarrow C$. Thus the integer $d_r \geq 0$ exists and is unique.

Proposition 4.5. For every $r \geq 0$,

$$d_r \leq \Delta_{\lfloor r/2 \rfloor}.$$

In particular, for every $m \geq 0$,

$$(13) \quad d_{2m} \leq 2m - s_2(m).$$

Proof. Put $m = \lfloor r/2 \rfloor$. Choose $c_r \in \pi_{2r}C$ whose image generates the torsion-free quotient and whose rational image is an odd unit times $2^{-d_r}u^r$. By (4), this element lifts uniquely to $\pi_{2r}F_m C$. After rescaling by an odd unit, the definition of d_r gives a torsion class t such that

$$2^{d_r}c_r - u^r = t.$$

Apply the natural map to $\mathcal{K}(F_m C)$ and invert 2. The torsion class vanishes, so c_r maps to $2^{-d_r}u^r$ in the common rational line (10). Therefore $2^{-d_r}u^r \in \Lambda_m$. Lemma 4.3 forces $d_r \leq \Delta_m$. Taking $r = 2m$ gives (13). □

5. TATE CONSTRUCTIONS OF THE FINITE POSTNIKOV SECTIONS

We express the Tate construction of each finite Postnikov section as a two-term Laurent-series cofiber and reduce the order of its unit to the denominator invariant of Definition 4.4.

Set

$$M_r = \tau_{\leq 2r}C.$$

The truncation is taken in connective commutative A -algebras, so $C \rightarrow M_r$ is unital. Only the underlying A -module is used in the next two lemmas.

Lemma 5.1. For every $r \geq 0$, the A -module M_r is perfect.

Proof. By Proposition 3.1, the homotopy groups of C are finitely generated $\mathbb{Z}_{(2)}$ -modules. The finite Postnikov tower of M_r has layers corresponding to finite sums of $\mathbb{Z}_{(2)}$ and $\mathbb{Z}/2^a$. In the heart of Mod_A , the positive-degree class u acts trivially, so these layers are finite sums of suspensions of

$$H\mathbb{Z}_{(2)} \simeq A/u \quad \text{and} \quad H\mathbb{Z}/2^a \simeq A/(u, 2^a).$$

Both are finite-cell A -modules. Perfect modules are closed under finite extensions, proving the claim. $\square$

Let x be a variable of homotopy degree -2 . We use the completed Laurent-series $A((x))$ of [AMS98, Section 1]; its coefficients are allowed to be infinite in the nonnegative powers of x but finite in the negative powers, and $\pi_* A((x)) = A_*((x))$.

Lemma 5.2 (The ku -Tate module). *Put*

$$Q_A = \text{cofib}(2 - ux : A((x)) \longrightarrow A((x))).$$

There is an equivalence of A -modules

$$(14) \quad \theta : Q_A \xrightarrow{\simeq} A^{tC_2}$$

for which $\theta_(\bar{1})$ is the multiplicative Tate unit, where $\bar{1} \in \pi_0 Q_A$ is represented by $1 \in \pi_0 A((x))$.*

Proof. Write $A_* = \pi_* A = \mathbb{Z}_{(2)}[u]$. Choose the multiplicative coordinate for the canonical complex orientation of $A = ku_{(2)}$. The hypothesis in Ando–Morava–Sadofsky that the 2-series be a non-zero-divisor holds: A_* is a domain, and the least-exponent argument below shows that multiplication by $2 - ux$ (and hence by $x(2 - ux)$) is injective. A choice of sign of x gives

$$[2](x) = x(2 - ux).$$

Multiplication by $2 - ux$ is injective on $A_*((x))$. Indeed, for a nonzero Laurent series with least x -exponent j , the coefficient of x^j after multiplication is twice its leading coefficient, which is nonzero because $A_* = \mathbb{Z}_{(2)}[u]$ is 2-torsion-free. Therefore

$$(15) \quad \pi_* Q_A \cong A_*((x))/(2 - ux).$$

Ando–Morava–Sadofsky compute, as a ring,

$$\pi_* A^{tC_2} \cong A_*((x))/([2](x))$$

in [AMS98, Lemma 2.1]. Since x is invertible, this coefficient module is the module in (15). In every degree,

$$(16) \quad \pi_{2d} Q_A \cong \mathbb{Z}_2\{x^{-d}\}, \quad \pi_{2d+1} Q_A = 0, \quad ux^{-d} = 2x^{-(d+1)}.$$

In particular, $\pi_0 Q_A \cong \mathbb{Z}_2$; the allowed infinite nonnegative powers of x account for the 2-adic completion. Under the resulting embedding in $\mathbb{Q}_2[u^{\pm 1}]$, the generator x^{-d} corresponds to $(u/2)^d$.

We use a separate realization argument to pass from coefficients to A -modules. Let $ku_{\mathbb{Z}}$ denote integral connective complex K -theory, so that

$A = (ku_{\mathbb{Z}})_{(2)}$. After restricting scalars along $ku_{\mathbb{Z}} \rightarrow A$, the graded module in (16) is the completed module Q_1 of Hu–Kriz–Somberg. Their general length-one realization statement, together with the completed construction immediately preceding Theorem 11, gives a unique $ku_{\mathbb{Z}}$ -module realization of Q_1 [HKS19, the discussion following Lemma 1 and the paragraph preceding Theorem 11]. At $n = 1$, the Q_1 -multiplicity in [HKS19, Theorem 11] is one and the Eilenberg–Mac Lane error term vanishes. Consequently,

$$\mathrm{Res}_{ku_{\mathbb{Z}}}^A Q_A \simeq ku_{\mathbb{Z}}^{tC_2}$$

as $ku_{\mathbb{Z}}$ -modules.

It remains to compare integral and 2-local connective K -theory. Let D be the cofiber of $ku_{\mathbb{Z}} \rightarrow A$. For $i \geq 0$,

$$\pi_{2i} D \cong \mathbb{Z}_{(2)}/\mathbb{Z},$$

and all other homotopy groups vanish. Multiplication by 2 is therefore an equivalence on D . Hence $D^{tC_2} \simeq 0$ [NS18, Lemma I.2.8]. Exactness of the Tate construction now gives a $ku_{\mathbb{Z}}$ -linear equivalence

$$ku_{\mathbb{Z}}^{tC_2} \xrightarrow{\simeq} \mathrm{Res}_{ku_{\mathbb{Z}}}^A A^{tC_2}.$$

Finally, $ku_{\mathbb{Z}} \rightarrow A$ is an idempotent localization, so $A \otimes_{ku_{\mathbb{Z}}} A \simeq A$ and restriction

$$\mathrm{Res}_{ku_{\mathbb{Z}}}^A : \mathrm{Mod}_A \longrightarrow \mathrm{Mod}_{ku_{\mathbb{Z}}}$$

is fully faithful. The preceding composite therefore corresponds, up to contractible choice, to an A -module equivalence $Q_A \simeq A^{tC_2}$.

Choose such an equivalence. On π_0 , it is an isomorphism between two copies of $\mathbb{Z}_2$. Hence $\bar{1}$ maps to an odd element z , which is a unit in $\pi_0(A^{tC_2})$. Postcomposing with the A -linear automorphism given by multiplication by z^{-1} produces (14) with $\theta_*(\bar{1}) = 1$. Thus AMS supplies the coefficient calculation, while HKS supplies the module realization. $\square$

Lemma 5.3 (Projection formula). *If M is a perfect A -module with trivial C_2 -action, then the canonical map*

$$M \otimes_A A^{tC_2} \longrightarrow M^{tC_2}$$

is an equivalence. If, in addition, M is a unital A -algebra, this comparison carries $1 \otimes 1$ to the Tate unit.

Proof. A perfect A -module is dualizable, so $M \otimes_A -$ preserves limits as well as colimits. It therefore commutes with homotopy fixed points and homotopy orbits of the constant C_2 -diagram on A :

$$M \otimes_A A^{hC_2} \simeq M^{hC_2}, \quad M \otimes_A A_{hC_2} \simeq M_{hC_2}.$$

These equivalences intertwine the norm maps. Taking cofibers proves the assertion. If M is a unital A -algebra, the comparison is induced by its action and the lax symmetric monoidal structure of Tate, so it carries $1 \otimes 1$ to the Tate unit. $\square$

For any A -module M , write $M((x)) = M \otimes_A A((x))$. Combining Lemmas 5.1, 5.2, and 5.3 gives

$$(17) \quad M_r^{tC_2} \simeq \text{cofib}(2 - ux: M_r((x)) \longrightarrow M_r((x))).$$

Because M_r is dualizable, tensoring with it commutes with the inverse limit defining $A((x))$. Since M_r is connective and $2r$ -truncated, the coefficient towers contributing in total degrees zero and one stabilize after finitely many terms. Thus there is no product or $\lim^1$ ambiguity in degree zero, and

$$(18) \quad \pi_0 M_r((x)) = \bigoplus_{j=0}^r \pi_{2j} C x^j.$$

The cofiber long exact sequence gives an injection

$$(19) \quad \text{coker}(2 - ux: \pi_0 M_r((x)) \rightarrow \pi_0 M_r((x))) \hookrightarrow \pi_0 M_r^{tC_2}.$$

The class represented by 1 is the Tate unit.

The following Smith-normal-form calculation estimates the order of the class represented by 1.

Lemma 5.4 (The unit-order lemma). *For every $r \geq 0$, put*

$$L_r = \pi_0 M_r((x)) / \text{tors}(\pi_0 M_r((x))).$$

Since $\pi_0 C = \mathbb{Z}_{(2)}$, we have $d_0 = 0$. The class of the unit in $\text{coker}(2 - ux: L_r \rightarrow L_r)$ has exact order

$$(20) \quad 2^{E_r}, \quad E_r = \max_{0 \leq j \leq r} (j + 1 - d_j).$$

Consequently, the order of the Tate unit in $\pi_0 M_r^{tC_2}$ is at least 2^{E_r} .

Proof. Take $c_0 = 1$, and for $j > 0$ choose a primitive generator c_j for the torsion-free quotient of $\pi_{2j} C$, with rational image an odd unit times $u^j / 2^{d_j}$. Multiplication by u maps the $(j-1)$ st integral lattice into the j th, so $d_j \geq d_{j-1}$ and, modulo torsion,

$$uc_{j-1} = \epsilon_j 2^{d_j - d_{j-1}} c_j, \quad \epsilon_j \in \mathbb{Z}_{(2)}^\times.$$

Put $X_j = c_j x^j$. Then $X_0, \dots, X_r$ form a $\mathbb{Z}_{(2)}$ -basis of L_r , and the cokernel $\text{coker}(2 - ux: L_r \rightarrow L_r)$ has the presentation

$$\begin{aligned} 2X_{j-1} &= \epsilon_j 2^{d_j - d_{j-1}} X_j & (1 \leq j \leq r), \\ 2X_r &= 0. \end{aligned}$$

The last relation occurs because $\pi_{2r+2} M_r = 0$.

To determine the least e for which $2^e X_0$ vanishes in the cokernel, solve

$$2^e X_0 = (2 - ux) \sum_{j=0}^r \lambda_j X_j$$

in $L_r \otimes_{\mathbb{Z}_{(2)}} \mathbb{Q}$. Comparing coefficients determines the solution uniquely:

$$\lambda_0 = 2^{e-1}, \quad \lambda_j = \epsilon_j 2^{d_j - d_{j-1} - 1} \lambda_{j-1}.$$

Hence

$$\nu_2(\lambda_j) = e - 1 + d_j - j.$$

All coefficients are integral if and only if $e \geq j + 1 - d_j$ for every j . This proves (20).

The torsion subgroup of (18) is preserved by $2 - ux$. Passing to the torsion-free quotient therefore induces a surjection on cokernels. Together with (19), this shows that the order computed above is a lower bound for the order of the Tate unit. $\square$

6. UNBOUNDED UNIT ORDERS AND THE COUNTEREXAMPLE

Lemma 5.4 and Proposition 4.5 give the finite-stage bound.

Proposition 6.1. *For every $m \geq 0$,*

$$(21) \quad \text{ord}(1 \in \pi_0((\tau_{\leq 4m} C)^{tC_2})) \geq 2^{s_2(m)+1}.$$

Here an infinite order is allowed on the left.

Proof. Apply Lemma 5.4 with $r = 2m$. The term $j = 2m$ in (20), together with Proposition 4.5, gives

$$E_{2m} \geq 2m + 1 - d_{2m} \geq 2m + 1 - (2m - s_2(m)) = s_2(m) + 1.$$

$\square$

Proof of Theorem 1.1. Proposition 2.1 proves $K(1) \wedge C \simeq 0$.

For $q \geq 1$, set $m = 2^q - 1$. Then $s_2(m) = q$, so Proposition 6.1 gives

$$\text{ord}(1 \in \pi_0((\tau_{\leq 4(2^q-1)} C)^{tC_2})) \geq 2^{q+1}.$$

The orders of these finite-stage Tate units are therefore unbounded.

Suppose for contradiction that $(C^{tC_2})_{\mathbb{Q}} \simeq 0$. Since $\pi_0(C^{tC_2})$ is 2-local, its unit is killed by some fixed power 2^a . The Postnikov truncation maps are unital maps of E_{∞} -rings

$$C \longrightarrow \tau_{\leq N} C.$$

For the finite group C_2 and the trivial actions used here, the Tate construction is lax symmetric monoidal [NS18, Theorem I.3.1]; hence these maps induce unital maps

$$C^{tC_2} \longrightarrow (\tau_{\leq N} C)^{tC_2}.$$

The same power 2^a would consequently kill every finite-stage unit, contradicting the preceding unbounded lower bounds. Thus $(C^{tC_2})_{\mathbb{Q}} \not\simeq 0$, or equivalently

$$H\mathbb{Q} \wedge C^{tC_2} \not\simeq 0.$$

$\square$

Remark 6.2 (Postnikov continuity). For a finite group, the Tate construction is Postnikov-continuous: [NS18, Lemma I.2.6(i)] gives

$$C^{tC_2} \simeq \operatorname{holim}_N (\tau_{\leq N} C)^{tC_2}.$$

The proof above does not require this equivalence. It uses only the functorial, unit-preserving maps from the global Tate construction to the finite-stage Tate constructions.

Remark 6.3 (Role of power operations). The proof uses only the uniform exponent in Proposition 4.1; no power-operation recurrence on the affine algebra is required.

REFERENCES

- [AMS98] Matthew Ando, Jack Morava, and Hal Sadofsky, *Completions of $\mathbb{Z}/(p)$ -Tate cohomology of periodic spectra*, *Geom. Topol.* **2** (1998), 145–174.
- [BMMS86] Robert R. Bruner, J. Peter May, James E. McClure, and Mark Steinberger, *H_∞ ring spectra and their applications*, *Lecture Notes in Mathematics*, vol. 1176, Springer-Verlag, Berlin, 1986.
- [BSY22] Robert Burklund, Tomer M. Schlank, and Allen Yuan, *The chromatic Nullstellensatz*, arXiv:2207.09929v1, 2022.
- [HKS19] Po Hu, Igor Kriz, and Petr Somberg, *Tate cohomology of connected k -theory for elementary abelian groups revisited*, *J. Homotopy Relat. Struct.* **14** (2019), no. 3, 749–772.
- [NS18] Thomas Nikolaus and Peter Scholze, *On topological cyclic homology*, *Acta Math.* **221** (2018), no. 2, 203–409.